

SiPM-Wall MIP Calibration from Wall–Plastic Coincidences

n_TOF EAR2 X17 campaign, run 224404 (July 15, 2026)

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Abstract

Using the official n_TOF processed data for run 224404, we reconstruct coincidences between the SiPM trigger walls (WAL) and the back plastic scintillator bars (PSS) of the four MX17 arms. Time-difference sidebands give a clean statistical separation of true coincidences from combinatorial background. Arms B and C show unambiguous MIP peaks in the wall SiPM channels at 29 mV to 39 mV (~ 960 – 1270 ADC), providing a per-channel MIP calibration. Arms A and D show no MIP population in coincidence; wall-only top–bottom coincidences and plastic-gain studies constrain the cause. We propose reading out the liquid scintillators and adjusting plastic PMT voltages.

1 Data and event selection

Run 224404 (13 GB, 3792 bunches: 1471 dedicated $\sim 8.5 \times 10^{12}$ ppp, 2321 parasitic $\sim 4.1 \times 10^{12}$ ppp) from `/eos/experiment/ntof/processing/official/done/`. Each tree stores hit-level pulse-shape-analysis results; amplitudes are converted to mV with the per-channel DAQ full scale (≈ 0.0305 mV/ADC count, 16 bit, ~ 2 V full scale).

The SiPM walls delivered no data for bunches 643–2212 (46% of beam-on bunches; hardware/DAQ cause unknown, plastics unaffected — flagged for follow-up). All results use the 1571 post-recovery beam-on bunches, and hits more than 0.1 ms after the γ -flash, where coincidence purity is highest.

2 Coincidence method

For every wall–plastic channel pair, the distribution of $\Delta t = t_{\text{wall}} - t_{\text{plastic}}$ shows a narrow peak (rms 1.5–3.5 ns after per-channel alignment) on a flat combinatorial pedestal. The pedestal is measured in the $|\Delta t| > 100$ ns sidebands and subtracted (Fig. 1); a ± 8 ns calibrated window is used for the signal sample, with an off-peak sideband sample subtracted from every downstream spectrum. Per-channel time offsets (-5 to -20 ns, up to 9 ns spread within an arm) are measured on the high-purity sample and stored as a calibration (`calib/time_offsets_run224404.json`).

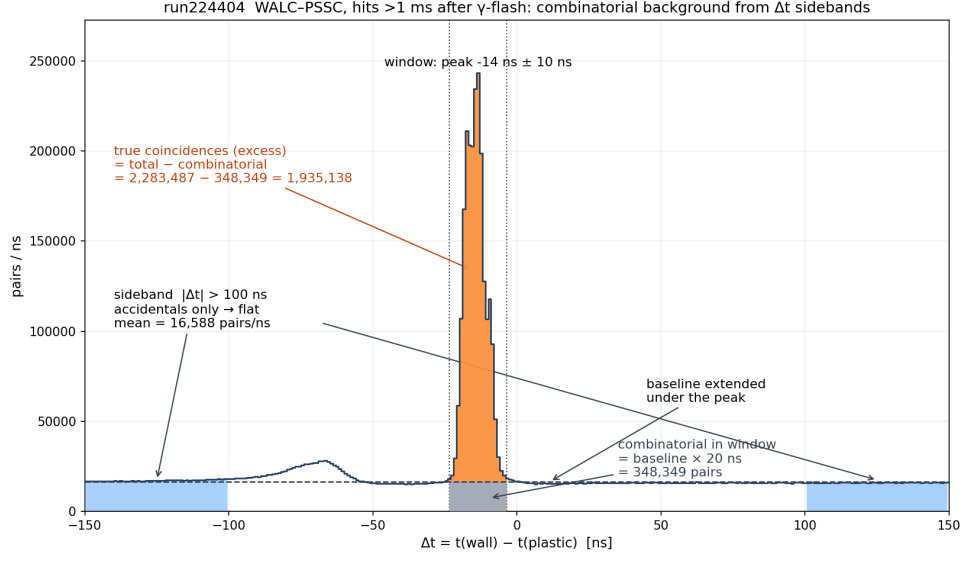


Figure 1: Combinatorial-background estimation on real data (WALC-PSSC, >1 ms after flash).

3 Detector mapping confirmed in data

Within each wall, the channel-channel coincidence matrix is block-diagonal in the pairs (1,2),(3,4),(5,6),(7,8): each pair is the top/bottom readout of one 4-bar group (2.2–2.6M excess pairs each, 52–91% of a channel’s coincidences; top–bottom Δt rms ~ 2.5 ns). Groups are ordered left→right (seen from the back) across the two plastic bars. Which slot is top remains *assumed* (odd = top) pending the cabling sheet. Cross-arm matrices show corner hotspots — WALA(7,8) $\times$ WALD(1,2) and WALB(1,2) $\times$ WALC(7,8) — identifying A–D and B–C as *adjacent* arms (ring order consistent with A→D→C→B).

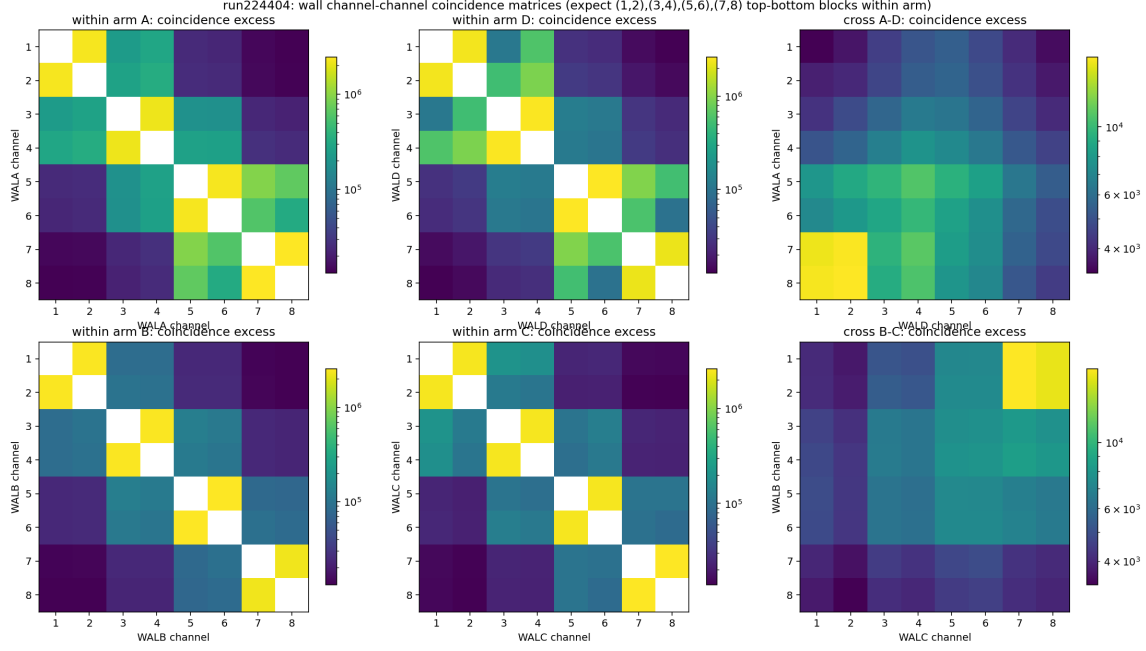


Figure 2: Wall channel-channel coincidence matrices: within-arm top/bottom blocks (left, middle) and cross-arm corner hotspots (right).

4 SiPM-wall MIP calibration (arms B, C)

The sideband-subtracted coincidence amplitude spectra of WALB/WALC show clean Landau-like MIP peaks in all 16 channels, well separated from threshold (Fig. 3). Peaks span 29.4 mV to 38.9 mV ($\sim 25\%$ channel-to-channel spread), giving direct per-channel MIP calibration constants. Note the physics of the selection: the plastic sits at the *back* of the arm, so the coincidence requires the track to traverse the wall and reach the plastic — a genuine through-going (MIP) selection for the *wall*. For the *plastic* itself the same selection only requires a hit, so the plastic spectra are hit spectra (including stopping/soft particles), not MIP spectra.

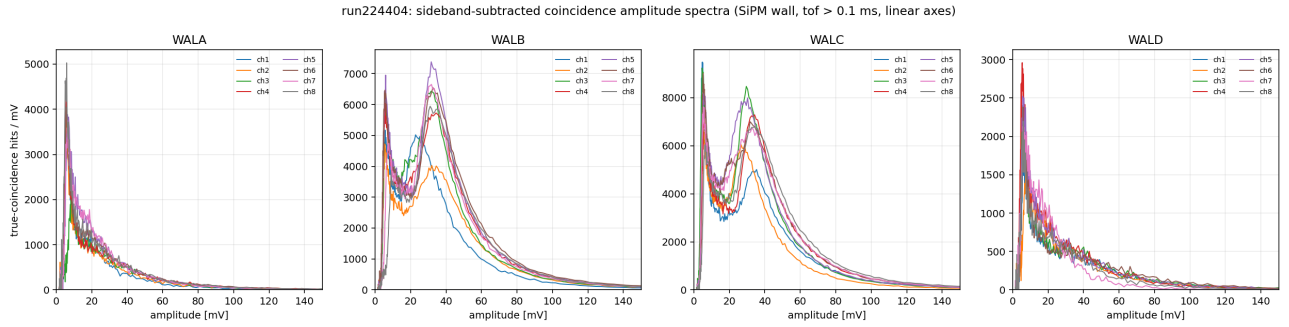


Figure 3: Sideband-subtracted wall coincidence amplitude spectra (linear, mV). B and C: clean MIP peaks; A and D: no MIP population.

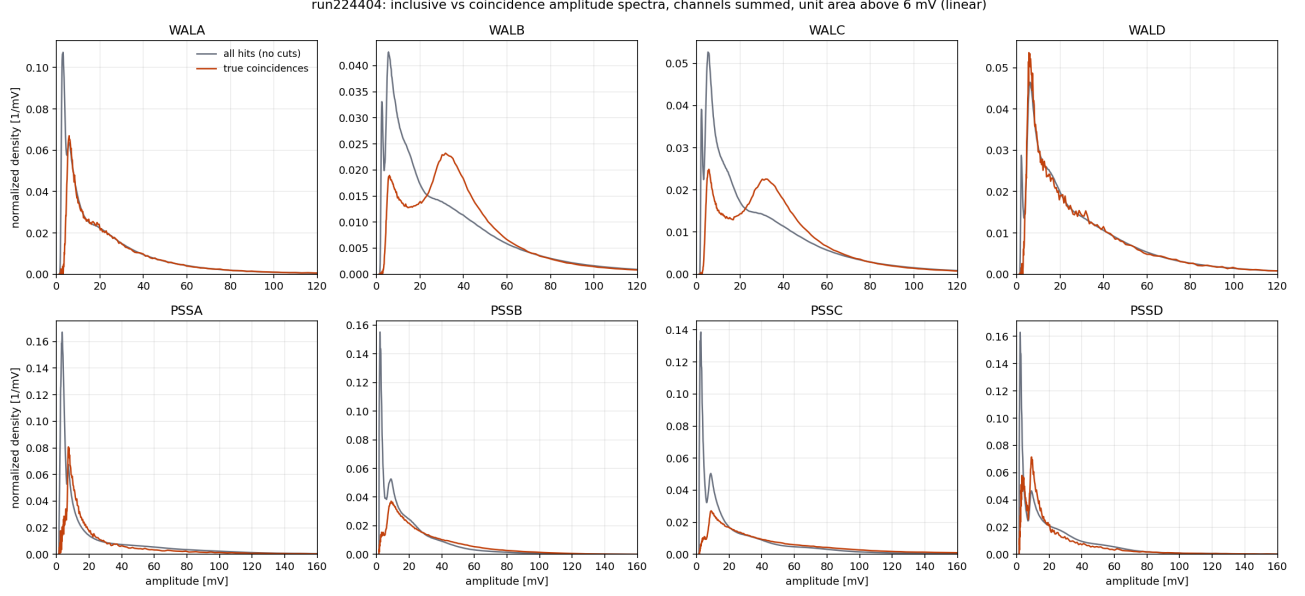


Figure 4: Inclusive vs coincidence-selected spectra (unit area): the coincidence isolates the MIP bump on B/C; on A/D it merely traces the inclusive shape.

5 The A/D problem

Arms A and D produce $6\text{--}8\times$ fewer true coincidences than B/C and their coincident wall spectra show *no* MIP bump — above $\sim 15\text{ mV}$ they simply trace the inclusive shape. Hypotheses: (i) plastic PMT gain/threshold too low on A/D, so MIP tracks fail to fire the plastic; (ii) wall SiPM gain low; (iii) geometry/orientation: A/D simply see less through-going flux. Inclusive plastic medians are within $\pm 30\%$ across all eight PMTs (PSSD1 highest), disfavoring a large plastic-gain deficit.

A wall-only cross-check — coincidences between the top and bottom SiPMs of the same 4-bar group (Fig. 5) — shows a MIP-scale bump at $25\text{--}40\text{ mV}$ on all four arms, so the SiPM response and gain are healthy everywhere, and every wall channel has an amplitude landmark. This check is *not* conclusive about the cause of the A/D deficit, however: top and bottom view the same scintillator, so a single localized deposit — including a γ Compton-scattering in the bars — lights both ends in tight coincidence. It therefore removes electronics noise but cannot distinguish through-going tracks from local deposits, and says nothing about whether particles continue to the back plastic. The A/D deficit remains unattributed among: plastic response, absorption between wall and plastic (e.g. if A/D carry the liquid-scintillator layers in front of the plastic — consistent with LIQA/LIQD being the cabled liquid channels, but unconfirmed), or a softer/more γ -rich flux at those arms. Discriminating follow-ups: the pre-flash cosmic sample (guaranteed through-going muons — do A/D wall-plastic coincidences appear at the expected rate?), the LIQ readout, and a plastic HV scan.

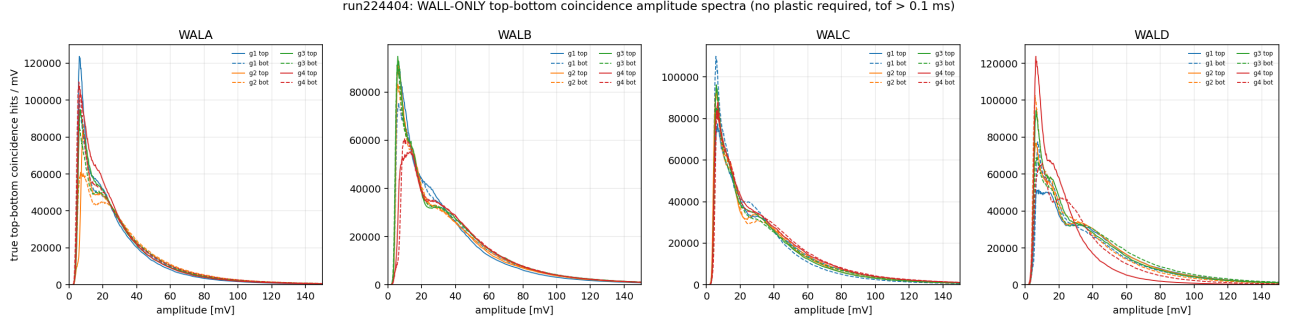


Figure 5: Wall-only top-bottom coincidence amplitude spectra (sideband subtracted): MIP-scale bump on all arms. Note top/bottom read the same bars, so this selects any real deposit, not through-going tracks.

6 Plastic PMT gains and HV suggestions

Relative gains from inclusive spectral medians (cross-arm comparable) with $G \propto V^7$:

PMT	HV [V]	median _{inc} [mV]	rel. gain	V (equalize)	V ($2\times$)
PSSA1 (AL)	1325	10.4	0.74	1384	1528
PSSA2 (AR)	1275	13.2	0.93	1288	1422
PSSB1 (BL)	1325	13.1	0.93	1339	1478
PSSB2 (BR)	1300	14.9	1.05	1290	1425
PSSC1 (CL)	1300	14.5	1.03	1295	1429
PSSC2 (CR)	1300	15.3	1.08	1285	1419
PSSD1 (DL)	1300	17.8	1.26	1258	1389
PSSD2 (DR)	1300	14.9	1.06	1290	1424

Table 1: Plastic PMT relative gains and HV suggestions ($\alpha = 7$ assumed; calibrate with a 2-point HV scan before trusting shifts >25 V).

7 Requests

1. **Read out the liquid scintillators** (LIQ; behind the plastic). A wall+LIQ coincidence requires passage *through* the plastic, isolating MIPs in the plastic and giving its true MIP calibration. Only LIQA/LIQD are cabled and neither is writing data.
2. Plastic HV scan (2 points suffice) to calibrate gain vs voltage, then apply the equalization above.
3. Confirm from cabling: top vs bottom SiPM card assignment; which 4 of the 20 wall bars are unread; physical positions of arms A–D.
4. Investigate the SiPM-wall outage (bunches 643–2212) with the DAQ crew.